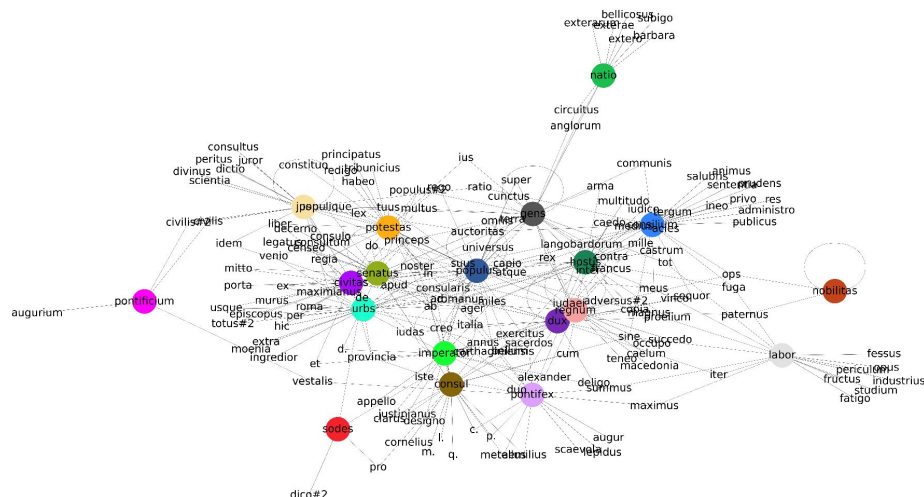


Investigating dimensions of lexical-semantic variation in Latin with distributional methods

Krzysztof Nowak and Barbara McGillivray

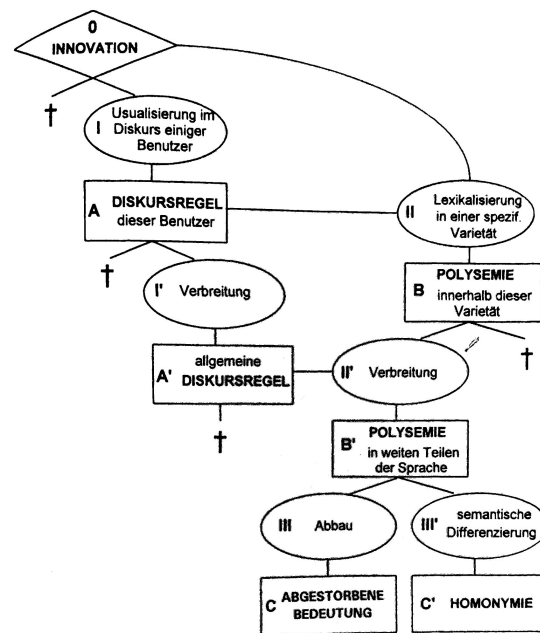
Language variation and meaning

- The central role of diachronic semantics
- Although existing academic dictionaries of Latin provide a rich description of a word's meaning based on complete evidence, they do not offer ready access to the semantic trajectory of the word, nor explicit quantitative information on the sense distribution in different authors, genres, or periods.
- Challenges for large-scale analyses
- We aim at investigating whether currently available corpora can form the basis for diachronic and variational studies of this sort



Semantic variation & change: *locus* and process

- senses are not equally spread across texts in corpora
- variation (examples: Adams 2007)
 - diatopic: *pullus* 'chicken' > 'cock'
 - diastratic: *quando* 'at which time, when'
 - diaphasic: *rostrum* 'beak' > '(human) face'
- semantic change happens at the level of discourse
 - "change does not originate within language ... but in language use" (Traugott & Dasher 2005)
 - "certain text types or subject matters may become the locus of innovations in certain semantic domains" (Traugott & Dasher 2005)
- stages
 - innovation by 1 or more speaker(s)
 - adoption as the rule of discourse: limited to a text type etc.
 - lexicalisation (↑ **polysemy**) / sense reduction (↓ **polysemy**)



(Blank 1997, p. 123)

Selected terms

- Polysemous socio-political terms (e.g. *ciuitas*, *populus*) referring to changing institutions of ancient and early medieval society
 - Amply attested in Latin texts through time
 - Their high frequency poses major challenges for manual analyses → perfect candidates for semi-automatic analyses
 - Previous research
- *civitas*
 - *consilium*
 - *senatus*
 - *hostis*
 - *imperator*
 - *natio*
 - *pontifex*
 - *potestas*

Meaning frequency and distribution in lexicography

- explicit markers

- frequency
 - labelling
- distribution
 - labelling

B notio loci, i. q. *urbs*, oppidum (i. aedificia sola, v. p. 1233, 63):
saepe de maiore urbe, ita ut locus, municipium, vicus, castellum sim.
opponantur (v. sub syn. sed cf. ISID. orig. 16, 2, 7 -es ... aut coloniae

cibus posuit, tanquam cibum poculis compensaret). **b** *apud* **ECCL.**
pars altaris tholo similis (cf. Ducange, II 324sq.): **GREG.** **TVR.** glor.

I *in imagine de contentione (praecipue apud ECCL. de mar-*
tyrio et pugna spirituali): **CIC.** de orat. 2, 317 in ipso illo

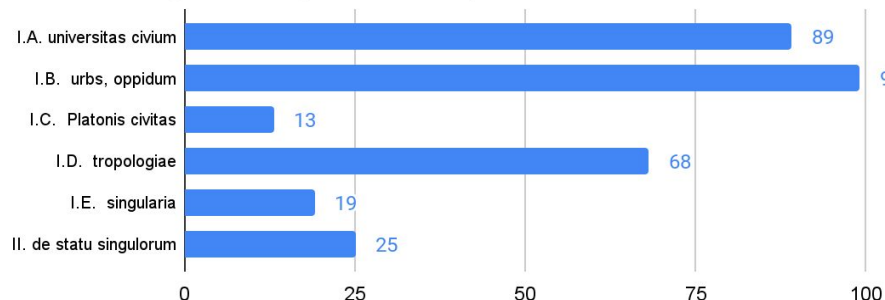
SVET. Iul. 25 **CENS.** 18, 11 p. 39, 15. **d** *multiplicatus pro sim-*
plicibus numeralibus apud poetas: bis -m: **VERG.** Aen. 8, 518

- implicit markers

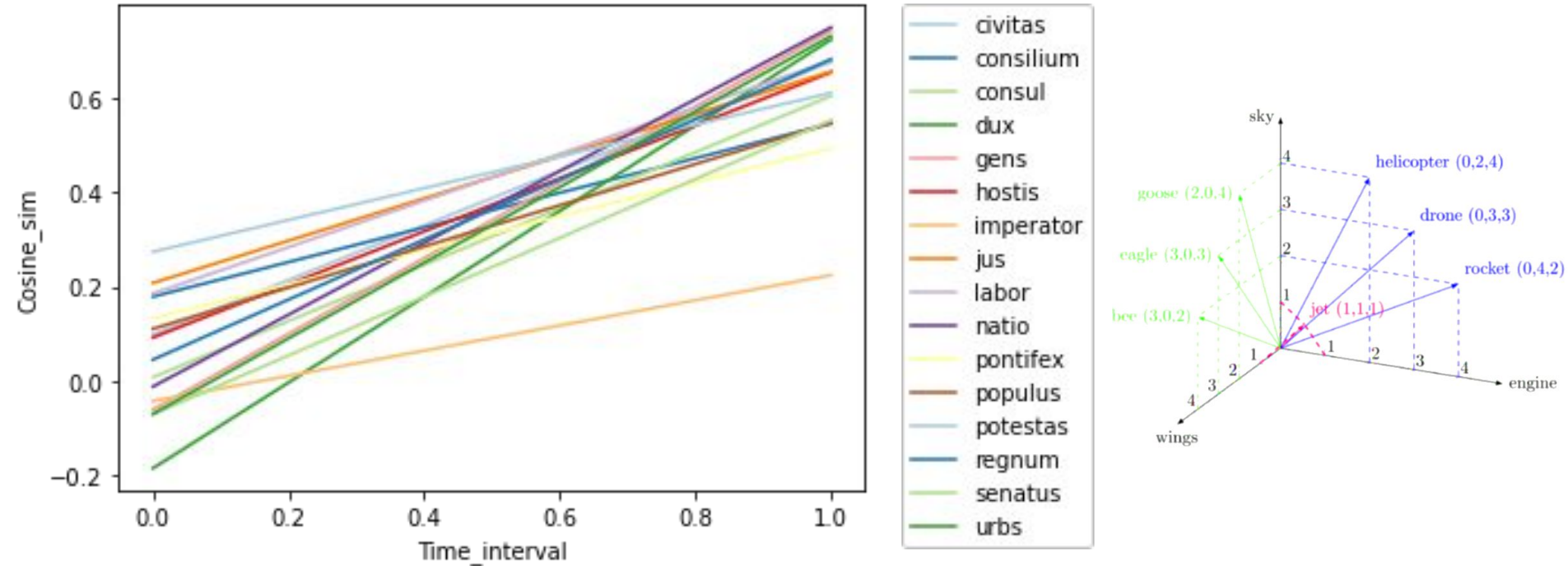
- frequency
 - number of lines
 - number of occurrences
- distribution
 - abbreviations

piamus eqs. **2** *latiore sensu fere i. q. lyra, fides sim. impi-*
mis apud poetas numero plurali (cf. PORPH. Hor. carm. 1, 17, 18 fides

CIVITAS. ThLL length of senses (in number of lines)



Semantic change analysis using word vectors



McGillivray, B. & Nowak, K. (2022). Tracing the semantic change of socio-political terms from Classical to early Medieval Latin with computational methods. In *Latin vulgaire – latin tardif XIV. 14th International Colloquium on Late and Vulgar Latin*.

Semantic change analysis using word vectors

Civitas: from citizenship to city

450BCE-1BCE	1BCE-450CE	450CE-900CE
gens, libertas, legatio, servitus, societas, ditio ,potens, potentia, princeps, status	urbs, oppidum, murus, porta, provincia, regio, domus, castellum, vasto, Italia	urbs, insula, villa, oppidum, castrum, regio, castellum, vicus, Gallia, sedes

McGillivray, B. & Nowak, K. (2022). Tracing the semantic change of socio-political terms from Classical to early Medieval Latin with computational methods. In *Latin vulgaire – latin tardif XIV. 14th International Colloquium on Late and Vulgar Latin*.

Dimensions of variation

- socio-linguistic features
 - place
 - **time**
- contextual features
 - **text type**
 - **genre**
 - register
 - topic
- LatinISE corpus (McGillivray and Kilgariff 2013)
- challenges
 - not all genres represented
 - no universally supported text classification etc.



The Perseus
Catalog



Metadata providers

Define the genre for the following Latin texts: A Firmo de civitate Dei, A solis ortus cardine, A tutte le genti per la inaugurazione del nuovo centro della Radio Vaticana in Santa Maria di Galeria, Ab Urbe condita, Academica, Ad D. D. Hannes, Familiares, Ad Galatam



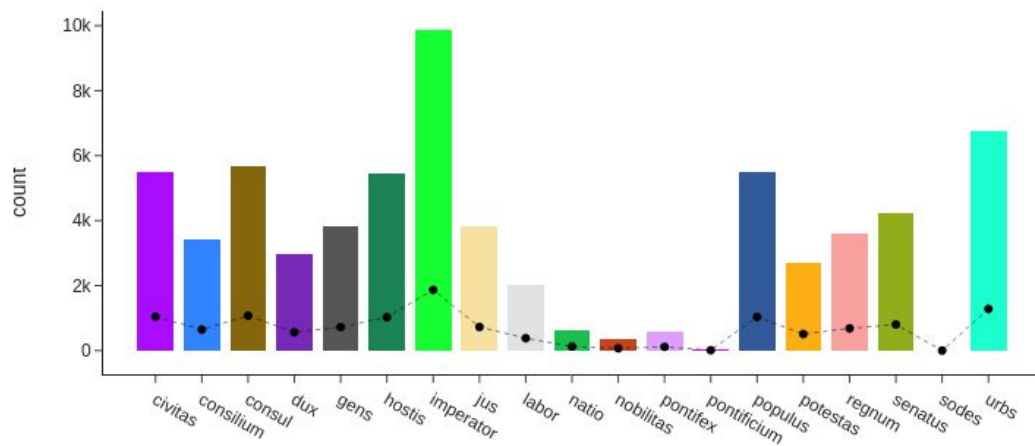
Here are the genres for the provided Latin texts:

1. A Firmo de civitate Dei - Philosophical treatise
2. A solis ortus cardine - Hymn
3. A tutte le genti per la inaugurazione del nuovo centro della Radio Vaticana in Santa Maria di Galeria - Speech
4. Ab Urbe condita - Historical work
5. Academica - Philosophical dialogue

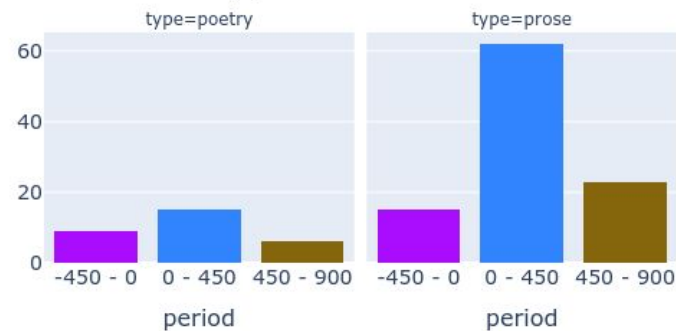
ChatGPT

Corpus, Text Types, and Periodization

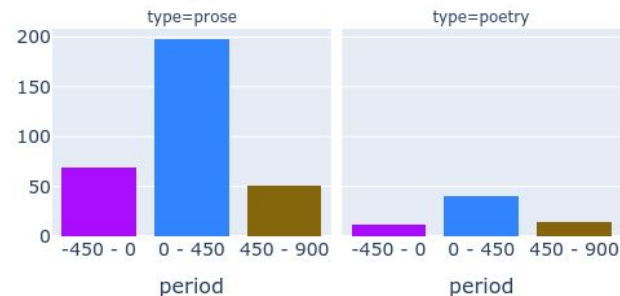
Period	Unique lemmas	Tokens	Unique tokens
-450 - 0	44861	1395858	103432
0 - 450	97396	2799762	195764
450 - 900	50265	1105116	97905
all	142051	5300736	262883



Number of texts



Number of unique authors



Analysis with embeddings

```
model.wv.similar_by_word('dies', 10)
```

```
[('hora', 0.6880184412002563),  
 ('mensis', 0.6146370768547058),  
 ('septimana', 0.608410537242889),  
 ('annus', 0.6053215265274048),  
 ('nox', 0.5529932975769043),  
 ('biduum', 0.5529341101646423),  
 ('triduum', 0.5428074598312378),  
 ('sabbatum', 0.5220836997032166),  
 ('vigilia', 0.5173648595809937),  
 ('Kalendae', 0.4951852858066559)
```

```
model.wv.similar_by_word('vir', 10)
```

```
[('Iohannes', 0.7155808806419373),  
 ('ovis', 0.6998714208602905),  
 ('mulier', 0.6878586411476135),  
 ('sanctus', 0.6756107211112976),  
 ('monasterium', 0.6755778193473816),  
 ('divitiae', 0.6590446829795837),  
 ('suscipio', 0.6576408743858337),  
 ('homo', 0.654313325881958),  
 ('frater', 0.6505364179611206),  
 ('beatus', 0.6500697135925293)]
```

Poetry v. prose: analysis with embeddings

Word	Freq prose	Freq poetry	Cosine	neighbours prose	neighbours poetry
imperator	9533	530	0.11	papa, Caesar, dux, imperium, divus, civis, comes, concedo, princeps, rex	pontifex, igitur, episcopus, ita, sacerdos, beo, quidam, diaconus, praedico, papa
ecclesiastical terms	Roman emperors and political life				
war and power					

imperator

—●— imperator, poetry
- - -◆- - imperator, prose



Poetry v. prose: analysis with embeddings

Word	Freq prose	Freq poetry	Cosine	neighbours prose	neighbours poetry
civitas	5379	308	0.49	urbs, regio, gens, arx, domus, monasterium, patria, ecclesia, populus, dux	iuxta, Ravenna, ingredior, infra, porta, praedico, sacerdos, episcopus, ecclesia, sepelio

urban
structure/locations

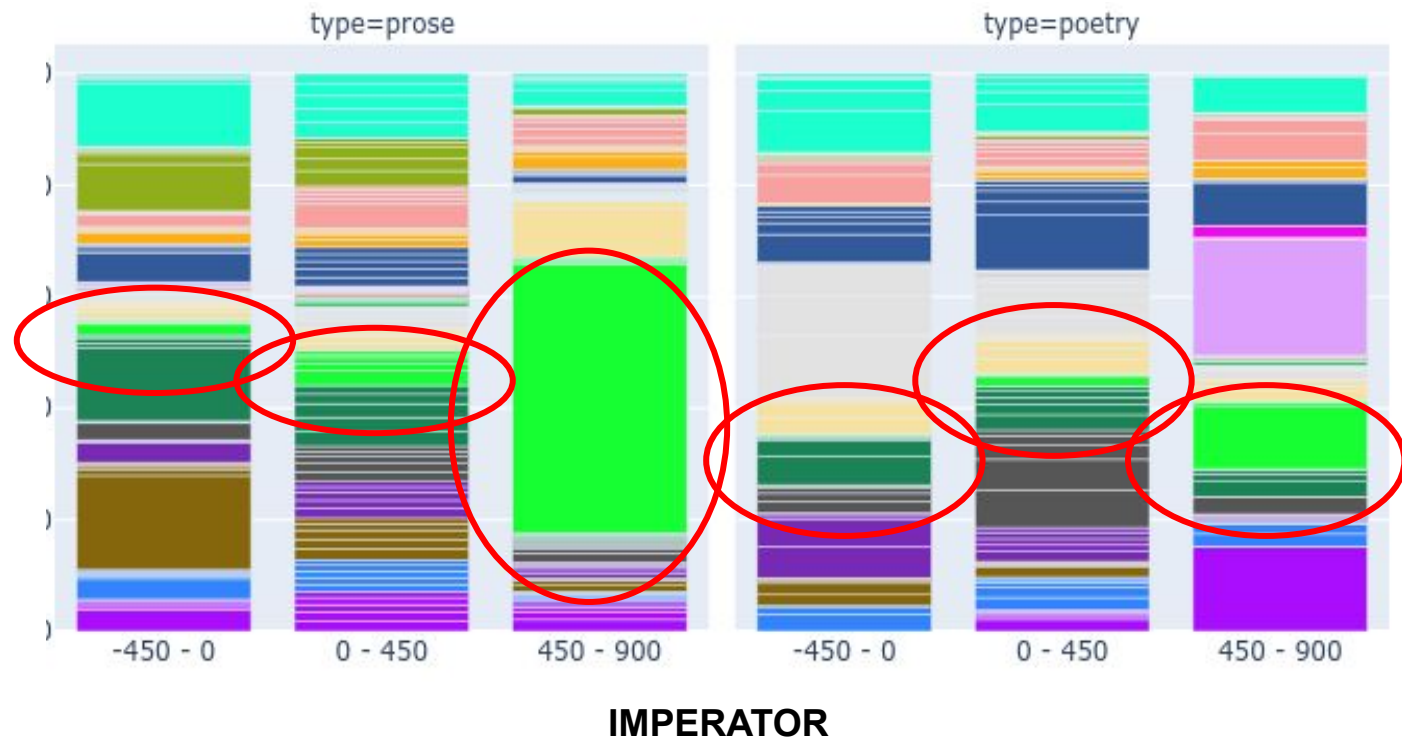
Church
organization

citizenship
civitas

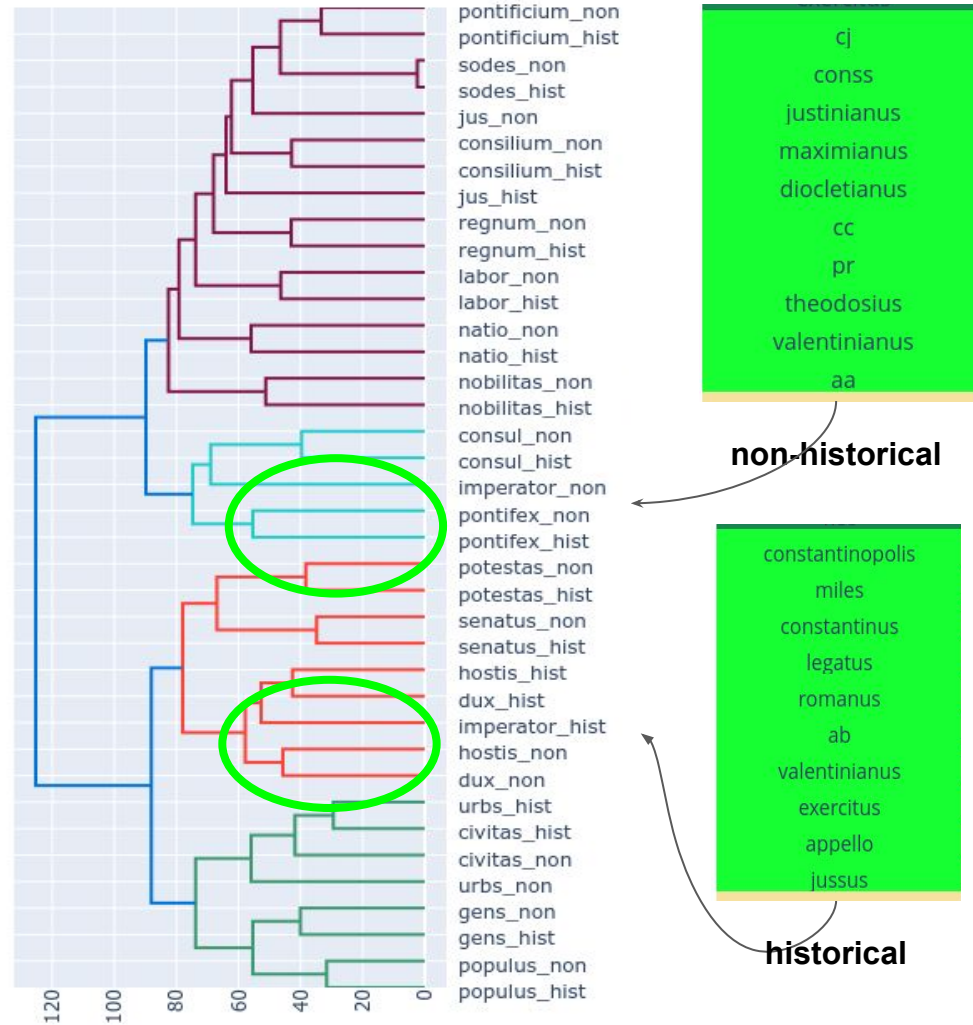
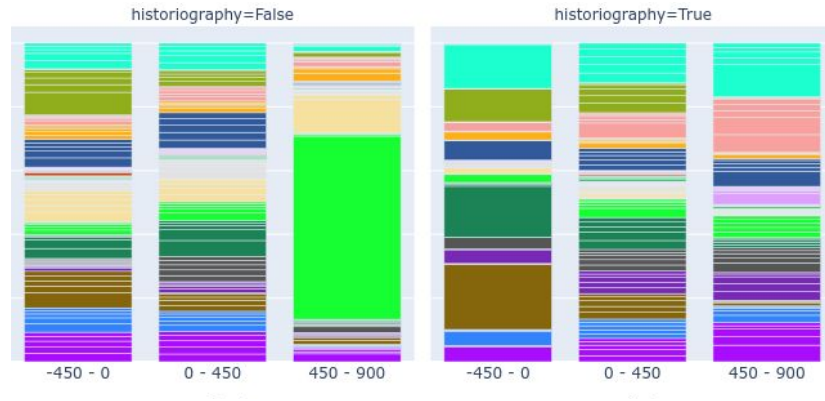
—●— civitas, poetry
- -◆- - civitas, prose



Poetry v. prose: key texts



Historical v. non-historical discourse



Conclusions

- large-scale automatic analysis needs to be combined with corpus-based study of word frequency
- the metadata for Latin texts require improvement
 - genre
 - register
 - topic
 - communicative features (Biber & Conrad)
- fixed-window data may not fit poetry well→ need for syntax-aware embeddings

Perspectives

- semantic change in onomasiological perspective: dynamics of semantic fields
- embeddings capture distributional relatedness rather than synonymy
- investigating semantic change in “discourses” (Blank)
 - topic domains
 - literary genres
 - technical writing
- selecting relevant variables
 - supporting lexicographic and linguistic study
 - choosing variables empirically